For the Use Only of a Registered Medical Practitioner

BESQUIL TABLETS
(Alprazolam Tablets USP 0.5 mg)

COMPOSITION

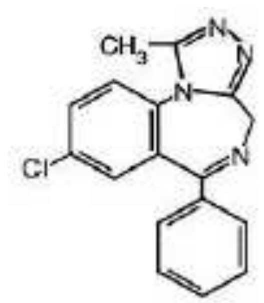
BESQUIL TABLETS 0.5 mg
Each tablet contains:
Alprazolam USP ………… 0.5 mg

PRODUCT DESCRIPTION

BESQUIL TABLETS 0.5 mg
Peach colored, circular, biconvex tablets having scoreline on one side.

DESCRIPTION

BESQUIL TABLETS contains alprazolam which is a triazolo analog of the 1, 4 benzodiazepine class of central nervous system-active compounds. The chemical name of alprazolam is 8-Chloro-1-methyl-6-phenyl-4H-s-triazolo [4, 3-l] [1, 4] benzodiazepine. The empirical formula is $C_{17}H_{13}ClN_4$ and the molecular weight is 308.76. The structural formula is:



ALPRAZOLAM

Alprazolam is a white crystalline powder, which is soluble in methanol or ethanol but which has no appreciable solubility in water at physiological pH.

PHARMACOLOGY

- **Mechanism of action**

CNS agents of the 1, 4 benzodiazepine class presumably exert their effects by binding at stereo specific receptors at several sites within the central nervous system. It is anxiolytic and mild antidepressant agent. Their exact mechanism of action is unknown. Clinically, all benzodiazepines cause a dose-related central nervous system depressant activity varying from mild impairment of task performance to hypnosis.

PHARMACOKINETICS

Following oral administration, alprazolam is readily absorbed. The peak plasma concentrations occur in 1-2 hours following administration. Plasma levels are proportional to the dose given; over the dose range of 0.5 to 3.0 mg, peak levels of 8.0 to 37 ng/mL are observed. Repeated dosage may lead to accumulation and this should be borne in mind in elderly patients and those with impaired renal or hepatic function. In vitro, alprazolam is bound (80 percent) to human serum protein. Serum albumin accounts for the majority of the binding. Using a specific assay methodology, the mean plasma elimination half-life of alprazolam is approximately 11.2 hours (range 6.3 – 26.9 hours) in healthy adults.

Alprazolam is extensively metabolized in humans, primarily by cytochrome P450 3A4 (CYP3A4), to two major metabolites in the plasma: 4-hydroxyalprazolam and α- hydroxyalprazolam. A benzophenone derived from alprazolam is also found in humans. Their half-lives appear to be similar to that of alprazolam. The plasma concentrations of 4-hydroxyalprazolam and α-hydroxyalprazolam relative to unchanged alprazolam concentration were always less than 4%. The reported relative potencies in benzodiazepine receptor binding experiments and in animal models of induced seizure inhibition are 0.20 and 0.66, respectively, for 4-hydroxyalprazolam and α-hydroxyalprazolam. Such low concentrations and the lesser potencies of 4-hydroxyalprazolam and α-hydroxyalprazolam suggest that they are unlikely to contribute much to the pharmacological effects of alprazolam. The benzophenone metabolite is essentially inactive. Alprazolam and its metabolites are excreted primarily in the urine.

Clinical studies in healthy volunteers have shown that single doses of alprazolam as high as 4 mg produce only effects which can be considered to be extensions of its pharmacological activity. No clinically significant effects on the cardiovascular or respiratory systems were observed. Sleep laboratory studies in man showed that alprazolam decreased sleep latency, increased duration of sleep and decreased the number of nocturnal awakenings.

Alprazolam produced small decreases in both stages 3-4 and REM sleep. Alprazolam did not affect the prothrombin times or plasma warfarin levels in male volunteers administered sodium warfarin orally.

In vitro, alprazolam is bound (80%) to human serum protein.

Pharmacokinetics in Special Populations

Changes in the absorption, distribution, metabolism and excretion of benzodiazepines have been reported in a variety of disease states including alcoholism, impaired hepatic function and impaired renal function.

Age: Changes in the absorption, distribution, metabolism and excretion of benzodiazepines have been demonstrated in geriatric patients. A mean half-life of alprazolam of 16.3 hours has been observed in healthy elderly subjects (range: 9.0 - 26.9 hours, n=16) compared to 11.0 hours (range: 6.3 - 15.8 hours, n=16) in healthy adult subjects. In patients with alcoholic liver disease, the half-life of alprazolam ranged between 5.8 and 65.3 hours (mean: 19.7 hours, n=17) as compared to between 6.3 and 26.9 hours (mean=11.4 hours, n=17) in healthy subjects. In an obese group of subjects, the half-life of alprazolam ranged between 9.9 and 40.4 hours (mean=21.8 hours, n=12) as compared to between 6.3 and 15.8 hours (mean=10.6 hours, n=12) in healthy subjects.

Gender — Gender has no effect on the pharmacokinetics of alprazolam.

Race — Maximal concentrations and half-life of alprazolam are approximately 15% and 25% higher in Asians compared to Caucasians.

Pediatric: The pharmacokinetics of alprazolam in pediatric patients have not been studied.

Cigarette Smoking — Alprazolam concentrations may be reduced by up to 50% in smokers compared to non-smokers.

INDICATIONS

BESQUIL TABLETS (Alprazolam tablets) is indicated for

Anxiety States (Anxiety Neuroses): Symptoms which occur in such patients include anxiety, tension, agitation, insomnia, apprehension, irritability and/or autonomic hyperactivity resulting in a variety of somatic complaints.

Mixed Anxiety-Depression: Symptoms of both anxiety and depression occur simultaneously in such patients.

Neurotic or Reactive Depression: Such patients primarily exhibit a depressed mood or a pervasive loss of interest or pleasure. Symptoms of anxiety, psychomotor agitation and insomnia are usually present. Other characteristics include appetite disturbances, changes in weight, somatic complaints, cognitive disturbances, decreased energy, feeling of worthlessness or guilt or thoughts of death or suicide.

Anxiety states, mixed anxiety-depression or neurotic depression associated with other diseases e.g., the chronic phase of alcohol withdrawal and functional or organic disease, particularly certain gastrointestinal, cardiovascular or dermatological disorders.

The effectiveness of **BESQUIL TABLETS** (Alprazolam tablets) for long-term use exceeding 6 months has not been established by systematic clinical trials. The physician should periodically reassess the usefulness of the drug for the individual patient.

DOSAGE AND ADMINISTRATION

The optimum dosage of **BESQUIL TABLETS** (Alprazolam tablets) should be individualized based upon the severity of the symptoms and individual patient response. The daily dosage (see table) will meet the needs of most patients. In the few patients who require higher doses, dosage should be increased cautiously to avoid adverse effects. When higher dosage is required, the evening dose should be increased before the daytime doses. In general, patients who have not previously received psychotropic medications will require lower doses than those previously treated with minor tranquilizers, antidepressants or hypnotics or those with a history of chronic alcoholism. It is recommended that the general principle of using the lowest effective dosage be followed to preclude the development of oversedation or ataxia. (See table.)

S.No	Indication	Usual Starting Dosage*	Usual Dosage Range
1	Anxiety	0.25 - 0.5 mg, given 3 times daily	0.5 - 4 mg daily, given in divided dose
2	Depression	0.5 mg, given 3 times daily	1.5 - 4.5 mg daily, given in divided

			dosage
3	Geriatric patients or in the presence of debilitating disease	0.25 mg, given 2-3 times daily	0.5 - 0.75 mg daily, given in divided dosage; to be gradually increased if needed and tolerated.
4	Panic-related disorders	0.5 - 1 mg given at bedtime	The dose should be adjusted to patient response. Dosage adjustments should be in increments no greater than 1 mg every 3 - 4 days. Additional doses can be added until 3 or 4 times daily schedule is achieved. The mean dose in a large multiclinical study was 5.7 ± 2.3 mg/day with rare patients requiring a maximum of 10 mg daily.
*If side effects occur, the dose should be lowered.			

WARNINGS AND PRECAUTIONS

Anaphylaxis (severe allergic reaction) and angioedema (severe facial swelling) which can occur as early as the first time the product is taken.

Complex Sleep - Related behaviors which may include sleep driving, making phone calls, preparing and eating food (while asleep).

Tolerance

Some loss of efficacy to the hypnotic effects of benzodiazepines may develop after repeated use for a few weeks.

Dependence

Use of benzodiazepines may lead to the development of physical and psychic dependence upon these products. The risk of dependence increases with dose and duration of treatment; it is also greater in patients with a history of alcohol and drug abuse.

Once physical dependence has developed, abrupt termination of treatment will be accompanied by withdrawal symptoms. These may consist of headaches, muscle pain, extreme anxiety, tension, restlessness, confusion and irritability. In severe cases the following symptoms may occur: derealization, depersonalisation, hyperacusis, numbness and tingling of the extremities, hypersensitivity to light, noise and physical contact, hallucinations or epileptic seizures.

Rebound insomnia and anxiety: a transient syndrome whereby the symptoms that led to treatment with a benzodiazepine recur in an enhanced form, may occur on withdrawal of treatment. It may be accompanied by other reactions including mood changes, anxiety or sleep disturbances and restlessness. Since the risk of withdrawal phenomena/rebound phenomena is greater after abrupt discontinuation of treatment, it is recommended that the dosage be decreased gradually by no more than 0.5 mg every three days. Some patients may require an even slower dose reduction.

Duration of treatment

The duration of treatment should bed as short as possible depending on the indication, but should not exceed eight to twelve weeks including tapering off process. Extension beyond these periods should not take place without re-evaluation of the situation.

It may be useful to inform the patient when treatment is started that it will be of limited duration and to explain precisely how the dosage will be progressively decreased. Moreover it is important that the patient should be aware of the possibility of rebound phenomena, thereby minimizing anxiety over such symptoms should they occur while the medicinal product is being discontinued.

There are indications, that in the case of benzodiazepines with a short duration of action, withdrawal phenomena can become manifest within the dosage interval, especially when the dosage is high. When benzodiazepines with a long duration of action are being used it is important to warn against changing to a benzodiazepine with a short duration of action, as withdrawal symptoms may develop.

Amnesia

Benzodiazepines may induce anterograde amnesia. The condition occurs most often several hours after ingesting the product and therefore to reduce the risk patients should ensure that they will be able to have uninterrupted sleep of 7-8 hours.

Psychiatric and 'paradoxical' reactions

Reactions like restlessness, agitation, irritability, aggressiveness, delusion, rages, nightmares, hallucinations, psychoses, inappropriate behaviour and other adverse behavioural effects are known to occur when using benzodiazepines. Should this occur, use of the drug should be discontinued.

They are more likely to occur in children and the elderly.

Specific patient groups

Pediatrics
Benzodiazepines should not be given to children without careful assessment of the need to do so; the duration of treatment must be kept to a minimum.

Elderly
The elderly should be given a reduced dose.

A lower dose is also recommended for patients with chronic respiratory insufficiency due to risk of respiratory depression.

Renal or Hepatic Insufficiency
Benzodiazepines are not indicated to treat patients with severe hepatic insufficiency as they may precipitate encephalopathy. Caution is recommended when treating patients with impaired renal or hepatic function.

Benzodiazepines are not recommended for the primary treatment of psychotic illness.

Benzodiazepines should not be used alone to treat depression of anxiety associated with depression (suicide may be precipitated in such patients). Administration to severely depressed or suicidal patients should be done with appropriate precautions and appropriate size of the prescription.

Benzodiazepines should be used with extreme caution in patients with a history of alcohol or drug abuse.

Laboratory Tests

Laboratory tests are not ordinarily required in otherwise healthy patients. However, when treatment is protracted, periodic blood counts, urinalysis, and blood chemistry analyses are advisable in keeping with good medical practice.

CONTRAINDICATIONS

BESQUIL TABLETS (Alprazolam) are contraindicated in patients with known sensitivity to this drug or other benzodiazepines or any of the other constituents of the tablet. **BESQUIL TABLETS** (Alprazolam) may be used in patients with open angle glaucoma who are receiving appropriate therapy, but is contraindicated in patients with acute narrow angle glaucoma.

BESQUIL TABLETS (Alprazolam) are contraindicated with ketoconazole and itraconazole, since these medications significantly impair the oxidative metabolism mediated by cytochrome P450 3A (CYP3A) (see WARNINGS and PRECAUTIONS–Drug Interactions).

BESQUIL TABLETS are also contraindicated in myasthenia gravis; severe respiratory insufficiency; sleep apnoea syndrome; severe hepatic insufficiency.

- **Pregnancy**

Teratogenic Effects: Pregnancy Category D.
Nonteratogenic Effects: It should be considered that the child born of a mother who is receiving benzodiazepines may be at some risk for withdrawal symptoms from the drug during the postnatal period. Also, neonatal flaccidity and respiratory problems have been reported in children born of mothers who have been receiving benzodiazepines. Due to the risk of congenital malformations associated with some minor tranquilizers, the use of Alprazolam should be avoided during pregnancy except when clearly needed and the anticipated benefit to the patient outweighs the potential risk to the fetus.

- **Lactation**

Benzodiazepines are known to be excreted in human milk. It should be assumed that alprazolam is as well. Chronic administration of diazepam to nursing mothers has been reported to cause their infants to become lethargic and to lose weight. As a general rule, nursing should not be undertaken by mothers who must use alprazolam tablets.

- **Labor and Delivery**

Alprazolam tablets have no established use in labor or delivery.

- **Pediatric Use**

Safety and effectiveness of alprazolam tablets in individuals below 18 years of age have not been established.

- **Geriatric Use**

Alprazolam should be given in reduced dose. A lower dose is also recommended for patients with chronic respiratory insufficiency due to risk of respiratory depression.

DRUG INTERACTIONS

Not recommended: Concomitant intake with alcohol

The sedative effects may be enhanced when the product is used in combination with alcohol. This affects the ability to drive or use machines.

Take into account: Combination with CNS depressants

Enhancement of the central depressive effect may occur in cases of concomitant use with antipsychotics (neuroleptics), hypnotics, anxiolytics/sedatives, antidepressant agents, narcotic analgesics, anti-epileptic drugs, anaesthetics and sedative antihistamines.

Pharmacokinetic interactions can occur when alprazolam is administered along with drugs that interfere with its metabolism. Compounds that inhibit certain hepatic enzymes (particularly cytochrome P450 3A4) may increase the concentration of alprazolam and enhance it's activity. Data from clinical studies with alprazolam, *in-vitro* studies with alprazolam and clinical studies with drugs metabolised similarly to alprazolam provide evidence for varying degrees of interaction and possible interaction with alprazolam for a number of drugs. Based on the degree of interaction and the type of data available, the following recommendations are made:

- The co-administration of alprazolam with ketoconazole, itraconazole, or other azole-type antifungals is not recommended.

- Caution and consideration of dose reduction is recommended when alprazolam is co-administered with nefazodone, fluvoxamine and cimetidine.

- Caution is recommended when alprazolam is co-administered with fluoxetine, propoxyphene, oral contraceptives, sertraline, diltiazem, or macrolide antibiotics such as erythromycin and troleandomycin.

- Interactions involving HIV protease inhibitors (e.g. ritonavir) and alprazolam are complex and time dependent. Low doses of ritonavir resulted in a large impairment of alprazolam clearance, prolonged its elimination half-life and enhanced clinical effects, however, upon extended exposure to ritonavir, CYP3A induction offset this inhibition. This interaction will require a dose-adjustment or discontinuation of alprazolam.

Drug/Laboratory Test Interactions

Although interactions between benzodiazepines and commonly employed clinical laboratory tests have occasionally been reported, there is no consistent pattern for a specific drug or specific test.

Carcinogenicity/ Mutagenicity/ Impairment of Fertility

No evidence of carcinogenic potential was observed during 2-year bioassay studies of alprazolam in rats at doses up to 30 mg/kg/day (150 times the maximum recommended daily human dose of 10 mg/day) and in mice at doses up to 10 mg/kg/day (50 times the maximum recommended daily human dose).

Alprazolam was not mutagenic in the rat micronucleus test at doses up to 100 mg/kg, which is 500 times the maximum recommended daily human dose of 10 mg/day. Alprazolam also was not mutagenic in vitro in the DNA Damage/Alkaline Elution Assay or the Ames Assay.

Alprazolam produced no impairment of fertility in rats at doses up to 5 mg/kg/day, which is 25 times the maximum recommended daily human dose of 10 mg/day.

ADVERSE REACTIONS

Sedation/drowsiness, light-headedness, numbed emotions, reduced alertness, confusion, fatigue, headache, dizziness, muscle weakness, ataxia, double or blurred vision, insomnia, nervousness/anxiety, tremor, change in weight. These phenomena occur predominantly at the start of therapy and usually disappear with repeated administration. Other side effects like gastrointestinal disturbances, changes in libido or skin reactions have been reported occasionally.

In addition, the following adverse events have been reported in association with the use of alprazolam: dystonia, anorexia, slurred speech, jaundice, musculoskeletal weakness, sexual dysfunction/changes in libido, menstrual irregularities, incontinence, urinary retention, abnormal liver function and hyperprolactinaemia. Increased intraocular pressure has been rarely reported.

Withdrawal symptoms have occurred following rapid decrease or abrupt discontinuance of benzodiazepines including alprazolam. These can range from mild dysphoria and insomnia to a major syndrome, which may include abdominal and muscle cramps, vomiting, sweating, tremor and convulsions. In addition, withdrawal seizures have occurred upon rapid decrease or abrupt discontinuation of therapy with alprazolam.

Amnesia

Anterograde amnesia may occur at therapeutic dosages, the risk increasing at higher dosages. Amnesic effects may be associated with inappropriate behaviour (see warnings and precautions).

Depression

Pre-existing depression may be unmasked during benzodiazepam use.

Psychiatric and 'paradoxical' reactions

Reactions like restlessness, agitation, irritability, aggressiveness, delusion, rages, nightmares, hallucinations, psychoses, inappropriate behaviour and other adverse behavioural effects are known to occur when using benzodiazepines or benzodiazepine-like agents. They may be quite severe with this product. They are more likely to occur in children and the elderly.

In many of the spontaneous case reports of adverse behavioural effects, patients were receiving other CNS drugs concomitantly and/or were described as having underlying psychiatric conditions. Patients who have borderline personality disorder, a prior history of violent or aggressive behaviour, or alcohol or substance abuse may be at risk of such events. Instances of irritability, hostility and intrusive thoughts have been reported during discontinuance of alprazolam in patients with post-traumatic stress disorder.

Dependence

Use (even at therapeutic doses) may lead to the development of physical dependence: discontinuation of the therapy may result in withdrawal or rebound phenomena (see warnings and precautions). Psychic dependence may occur. Abuse of benzodiazepines has been reported.

DRUG ABUSE AND DEPENDENCE

Controlled Substance Class

Alprazolam is a controlled substance under the Controlled Substance Act by the Drug Enforcement Administration and alprazolam tablets have been assigned to Schedule IV.

Physical and Psychological Dependence

Withdrawal symptoms similar in character to those noted with sedative/hypnotics and alcohol have occurred following discontinuance of benzodiazepines, including alprazolam tablets. The symptoms can range from mild dysphoria and insomnia to a major syndrome that may include abdominal and muscle cramps, vomiting, sweating, tremors and convulsions. Distinguishing between withdrawal emergent signs and symptoms and the recurrence of illness is often difficult in patients undergoing dose reduction. The long-term strategy for treatment of these phenomena will vary with their cause and the therapeutic goal. When necessary, immediate management of withdrawal symptoms requires re-institution of treatment at doses of alprazolam tablets sufficient to suppress symptoms. There have been reports of failure of other benzodiazepines to fully suppress these withdrawal symptoms. These failures have been attributed to incomplete cross-tolerance but may also reflect the use of an inadequate dosing regimen of the substituted benzodiazepine or the effects of concomitant medications.

While it is difficult to distinguish withdrawal and recurrence for certain patients, the time course and the nature of the symptoms may be helpful. A withdrawal syndrome typically includes the occurrence of new symptoms, tends to appear toward the end of taper or shortly after discontinuation, and will decrease with time. In recurring panic disorder, symptoms similar to those observed before treatment may recur either early or late, and they will persist.

While the severity and incidence of withdrawal phenomena appear to be related to dose and duration of treatment, withdrawal symptoms, including seizures, have been reported after only brief therapy with alprazolam tablets at doses within the recommended range for the treatment of anxiety (e.g., 0.75 to 4 mg/ day). Signs and symptoms of withdrawal are often more prominent after rapid decrease of dosage or abrupt discontinuance. The risk of withdrawal seizures may be increased at doses above 4 mg/day (see WARNINGS).

Patients, especially individuals with a history of seizures or epilepsy, should not be abruptly discontinued from any CNS depressant agent, including alprazolam. It is recommended that all patients on alprazolam tablets who require a dosage reduction be gradually tapered under close supervision.

Psychological dependence is a risk with all benzodiazepines, including alprazolam tablets. The risk of psychological dependence may also be increased at doses greater than 4 mg/day and with longer term use, and this risk is further increased in patients with a history of alcohol or drug abuse. Some patients have experienced considerable difficulty in tapering and discontinuing from alprazolam tablets, especially those receiving higher doses for extended periods. Addiction-prone individuals should be under careful surveillance when receiving alprazolam tablets. As with all anxiolytics, repeat prescriptions should be limited to those who are under medical supervision.

OVERDOSAGE

Manifestations of alprazolam overdosage include extensions of its pharmacologic activity, namely ataxia and somnolence, confusion, impaired coordination, diminished reflexes and coma. Death has been reported in association with overdoses of alprazolam by itself, as it has with other benzodiazepines. In addition, fatalities have been reported in patients who have overdosed with a combination of a single benzodiazepine, including alprazolam, and alcohol; alcohol levels seen in some of these patients have been lower than those usually associated with alcohol-induced fatality.

Overdosage reports with alprazolam tablets are limited. As in all cases of drug overdosage, respiration, pulse rate, and blood pressure should be monitored. General supportive measures should be employed, along with immediate gastric lavage, vomiting. Intravenous fluids should be administered and an adequate airway maintained. If hypotension occurs, it may be combated by the use of vasopressors. Dialysis is of limited value. As with the management of intentional overdosing with any drug, it should be borne in mind that multiple agents may have been ingested.

Flumazenil, a specific benzodiazepine receptor antagonist, is indicated for the complete or partial reversal of the sedative effects of benzodiazepines and may be used in situations when an overdose with a benzodiazepine is known or suspected. Prior to the administration of flumazenil, necessary measures should be instituted to secure airway, ventilation and intravenous access. Flumazenil is intended as an adjunct to, not as a substitute for, proper management of benzodiazepine overdose. Patients treated with flumazenil should be monitored for re-sedation, respiratory depression, and other residual benzodiazepine effects for an appropriate period after treatment. **The prescriber should be aware of a risk of seizure in association with flumazenil treatment, particularly in long-term benzodiazepine users and in cyclic antidepressant overdose.** The complete flumazenil package insert including CONTRAINDICATIONS, WARNINGS and PRECAUTIONS should be consulted prior to use.

STORAGE: Store below 30 °C, protected from light and moisture.

KEEP ALL MEDICINES OUT OF THE REACH OF CHILDREN.

SUPPLY
Blister pack of 10 x 10's

DATE OF REVISION
May 2014

MARKETING AUTHORIZATION HOLDER/ MANUFACTURER:
RANBAXY (MALAYSIA) SDN. BHD.
LOT 23, BAKAR ARANG INDUSTRIAL ESTATE
08000 SUNGAI PETANI
KEDAH, MALAYSIA